Armadale Kalamunda Group

Blood Glucose Monitoring – Adult Inpatient Guideline

1. Purpose

The purpose of this guideline is to provide clinical staff with guidance toward the requirements for blood glucose monitoring for adult inpatients.

2. Applicability

This guideline applies to all nursing and medical staff working across Armadale Kalamunda Group.

3. Requirements

3.1 Frequency of monitoring

The need to perform blood glucose level (BGL) monitoring on patients will be determined by the patient's clinical presentation or as directed by a Medical Officer or the Clinical Nurse Specialist – Diabetes.

The minimum blood glucose level (BGL) monitoring frequency for patients presenting with a primary diagnosis or co-morbidity of diabetes, will be on admission and then Standard Blood Glucose Level (BGL) Frequency; that is prior to breakfast, lunch, dinner and at 2100 hours.

Variances on the minimum monitoring frequency will be dependent upon the patient's condition as shown in the table below:

Patient status	Frequency of BGL Monitoring
Type 1 Diabetes	• Standard BGL frequency • If admitted with hypoglycaemia then additional 0200 test required • Review frequency within 24 hours of admission
Type 2 Diabetes OHA's and diet	• Standard BGL frequency • Review frequency within 24 hours of admission
Type 2 Diabetes insulin requiring	• Standard BGL frequency • If admitted with hypoglycaemia then additional 0200 test required • Review frequency within 24 hours of admission

Patient status	Frequency of BGL Monitoring
IV Insulin infusion	• Hourly as per DKA protocol
Long term care/rehabilitation	• Monitoring frequency to be individualised and based on goals of care
Fasting for procedure/investigation	• Standard BGL frequency
No diabetes (elevated BGL can occur in people without diabetes)	• Individualised based upon reason for hospitalisation and treatment • Liaise with MO to determine monitoring regime and target BGL

3.2 Additional monitoring

Ketone testing (urine and/or blood) is to be performed for:

- All Type 1 diabetic on admission.
- All patients on admission whose blood glucose level is greater than 15mmol/L.

If ketones are detected, continue ketones testing at the same frequency as blood glucose testing until no ketones are detected.

If Ketones are detected notify the medical officer and refer to the DKA protocol regarding haemodynamic monitoring and electrolyte level assays.

4. Documentation

- The Medical Officer will document the frequency of blood glucose testing on the Insulin Subcutaneous Order and Blood Glucose Record – Adult AKMR 174.1. This combined chart is used to document all blood glucose levels whether the patient is prescribed hypoglycaemics or diet controlled.
- Nursing staff will document the frequency of BGL monitoring in the Nursing Care Plan AKMR121A

5. Escalation

- For blood glucose levels outside target range, follow the prompts on the Insulin Subcutaneous Order and Blood Glucose Record – Adult AKMR174.1.

6. Procedure

6.1 Considerations

- Consider cultural, ethical and communication requirements
- Explain the procedure/s to the patient, family and/or carer and gain appropriate consent
- Maintain standard safety precautions for infection control

6.2 Preparation

- Clean area prior to testing and dry thoroughly (do not use an alcohol wipe/hand sanitiser as it may affect the result).
- Insert test strip into Glucometer
- Check expiry date and lot number of strip matches that displayed on monitor and if numbers do not correlate open a new box of test strips and calibrate the meter

6.3 Testing

- Using lancet device
- Place lancet firmly onto selected site, (Use the upper side of the finger) press the firing button
- Needle retracts into device immediately after firing
- Dispose into sharps container
- Massage from the base of the finger to assist blood flow.
- Hold sensor strip to blood sample and blood will be drawn into the strip
- Apply gentle pressure using cotton ball/ tissue to puncture site
- Wipe over Glucometer after use with large isopropyl 70% alcohol wipes
- Results will be displayed on the LCD screen

6.4 Documentation

- Document Blood Glucose Level on the Insulin Subcutaneous Order and Blood Glucose Record – Adult AKMR 174.1. Follow the ALERTS on this chart if BGL outside of Target Range

7. Legislative context

The following documents are required to give effect to this policy:

- Health Department of Western Australia (HDWA) WA Health High Risk Medication Policy MP 0131/20
- East Metropolitan Health Service (EMHS) Consent EMHS: 5
- AKG Recognition and Response to Acute Deterioration Policy

8. Supporting information

The following documents support the implementation of this policy:

- AKG Medication Management Procedure
- AKG Interpreter and Translator Guideline

9. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated via

- Monitoring of reported clinical incident data via Datix CIMS.

Department specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunity for improvement.

10. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.3 Healthcare rights and informed consent
- Standard 2 – Action 2.8 Communication that supports effective partnerships
- Standard 4 – Action 4.3 Partnering with consumers
- Standard 4 – Action 4.10 Medication review
- Standard 4 – Action 4.15 High-risk medicines
- Standard 8 – Action 8.6 Escalating care

11. References

1. NDDS.(2022).Blood Glucose Monitoring. National Diabetes Services Scheme. Blood glucose monitoring fact sheet ***
2. Australian Diabetes Educators Association (updated 2016). Sick Day Management. https://www.adea.com.au/events/webinar/sick-day-management/

12. Glossary

The following definitions are relevant to this policy.

Term	Definition
Type 1 Diabetes	Type 1 diabetes is an autoimmune condition this occurs when the immune system destroys the beta cells that produce insulin. Hence the pancreas can’t produce insulin.
Type 2 Diabetes	Type 2 diabetes is a progressive condition and occurs when the pancreas produces insufficient insulin accompanied by cellular resistance to insulin.
Standard BGL Frequency	Pre-meals and 2100.

13. Document owner

Director Nursing and Midwifery

Enquiries relating to this guideline may be directed to:

Title: Clinical Nurse Specialist Diabetes Educator
Department: Medical Services
Email: melanie.flynn@health.wa.gov.au

14. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
V4	04/06/2025	04/06/2025	Scheduled review

15. Authorisation

This guideline has been authorised and issued by the Director Nursing and Midwifery.

Authorised by	Sam Jenaway – A/Deputy Director Nursing and Midwifery
Authorisation date	04/06/2025
Published date	04/06/2025